

V04 FACILITIES FOR ACTIVE OCCUPANTS | O (MAX : 3 PT)

Intent : Foster active commuting through facilities that support cycling to the building and active occupants more broadly.

Summary : This WELL feature requires projects to provide no-cost bike storage along with showers, changing facilities and lockers, which support both active commuters and active occupants.

Issue : Cycle commuting is associated with lower cardiovascular disease incidence and mortality, cancer incidence and all-cause mortality.¹ From 2000-2016, federal funding in the U.S. for pedestrian and cyclist infrastructure increased from 0.1% to 2.2% of transportation spending, which translated to an increase in those walking and biking to work.² However, research suggests that many communities still lack sufficient funding and infrastructure to support active commuting intentions.² For example, among safety concerns and pervasive societal norms of the car culture, lack of bike parking and post-commute facilities are key reasons why people opt not to cycle to the workplace.³⁻⁸

Solutions : In a comprehensive review of the literature on this topic, studies reported that availability of amenities, such as bike parking and showers had a positive impact on cycling.⁹ Projections from one study showed that compared to baseline (5.8%), outdoor parking would increase those cycling to work to 6.3%, indoor, secure parking to 6.6% and indoor parking plus showers to 7.1%.¹⁰ In addition, lockers and changing/shower facilities, in particular, support activity goals and behaviors not only for cyclists but all occupants, such as those who might engage in physical activity or exercise before work. Such amenities signal to occupants that physical activity and, in particular, active commuting, is welcomed and encouraged. Thoughtful site planning and selection can also enhance opportunities for cycling. The presence of cyclist infrastructure, such as cyclist lanes and, in particular, infrastructure that promotes cyclist safety, is known to increase ridership.¹¹⁻¹³

PART 1 PROVIDE CYCLING INFRASTRUCTURE (MAX : 2 PT)

For All Spaces except Dwelling Units & Retail Spaces:

1: Cycling network

One of the following requirements is met:

- a. The project's address achieves a minimum Bike Score® of 50.¹⁴ To utilize this tool, projects must be located in a country that is "supported" by the tool.
- b. The building is within a 200 m walk distance of an existing cycling network that connects riders to at least 10 use types that are within a 4.8 km cycling distance of the project boundary.¹⁵ Uses and restrictions are defined in Appendix V1.¹⁶
- c. The project demonstrates existing plans for a cycling network that meets requirements a or b.

2: Bike parking

The following requirements are met:

- a. Bike parking is provided to occupants at no cost in the following quantities:
 1. Short-term bike parking (e.g., public bike rack) is located within a 30 m walk distance of a functional building entrance and can accommodate at least 2.5% of peak visitors (minimum of four spaces per building).¹⁵
 2. Long-term bike parking (e.g., bike room) is available within the project boundary and can accommodate at least 5% of regular occupants, excluding occupants under eight years old (minimum of four spaces per building).¹⁵
- b. The project provides no cost access to basic bike maintenance tools (e.g., bike pump and patch kit) co-located with long-term bike parking or quarterly on-site bike maintenance services.

Note : Interiors projects may count base building amenities towards feature requirements.

For Retail Spaces:

1: Cycling network

One of the following requirements is met:

- a. The project's address achieves a minimum Bike Score® of 50.¹⁴ To utilize this tool, projects must be located in a country that is "supported" by the tool.
- b. The building is within a 200 m walk distance of an existing cycling network that connects riders to at least 10 use types that are within a 4.8 km cycling distance of the project boundary.¹⁵ Uses and restrictions are defined in Appendix V1.¹⁶
- c. The project demonstrates existing plans for a cycling network that meets requirements a or b.

2: Bike parking

The following requirements are met:

- a. Bike parking is provided to occupants at no cost in the following quantities:
 1. Short-term bike parking (e.g., public bike rack) is located within a 30 m walk distance of a functional building entrance and includes at least two short-term bike storage spaces per 465 m² of floor area (minimum of two

spaces per building).¹⁷

2. Long-term bike parking (e.g., bike room) is available within the project boundary and can accommodate at least 5% of regular occupants (minimum of two spaces per building).¹⁷
- b. The project provides no cost access to basic bike maintenance tools (e.g., bike pump and patch kit) co-located with long-term bike parking or quarterly on-site bike maintenance services.

Note : Interiors projects may count base building amenities towards feature requirements.

For Dwelling Units:

1: Cycling network

One of the following requirements is met:

- a. The project's address achieves a minimum Bike Score® of 50.¹⁴ To utilize this tool, projects must be located in a country that is "supported" by the tool.
- b. The building is within a 200 m walk distance of an existing cycling network that connects riders to at least 10 use types that are within a 4.8 km cycling distance of the project boundary.¹⁵ Uses and restrictions are defined in Appendix V1.¹⁶
- c. The project demonstrates existing plans for a cycling network that meets requirements a or b.

2: Bike parking

The following requirements are met:

- a. Bike parking is provided to occupants at no cost in the following quantities:
 1. Short-term bike parking (e.g., public bike rack) is located within a 30 m walk distance of a functional building entrance and can accommodate at least 2.5% of peak visitors (minimum of four spaces per building).¹⁵
 2. Long-term bike parking (e.g., bike room) is located within the project boundary and can accommodate at least 30% of regular occupants (minimum of one space per building).¹⁵
- b. The project provides no cost access to basic bike maintenance tools (e.g., bike pump and patch kit) co-located with long-term bike parking or quarterly on-site bike maintenance services.

Note : Interiors projects may count base building amenities towards feature requirements.

PART 2 PROVIDE SHOWERS, LOCKERS AND CHANGING FACILITIES (MAX : 1 PT)

For All Spaces except Dwelling Units & Guest Rooms:

The following requirements are met:

- a. Showers with changing facilities are available to occupants at no cost in a quantity listed below within a 200 m walk distance of the project boundary.¹⁵

Regular Occupants (age 12 or older)	Required Number of Showers
0 - 100	One
101 - 999	One plus one for every 150 occupants above 100 Calculation: $1 + (\# \text{ of occupants} - 100) / 150$
1,000 – 4,999	Eight plus one for every 500 occupants above 1,000 Calculation: $8 + (\# \text{ of occupants} - 1,000) / 500$
5,000 + occupants	16 plus one for every 1,000 occupants above 5,000 Calculation: $16 + (\# \text{ of occupants} - 5,000) / 1,000$

- b. At least five lockers are available for every shower provided by the project. Lockers are co-located (i.e., in the same area/room, not on separate floors) with shower facilities.

Note : Interiors projects may count base building amenities towards feature requirements.

REFERENCES

1. Celis-Morales CA, Lyall DM, Welsh P, et al. Association between active commuting and incident cardiovascular disease, cancer, and mortality: Prospective cohort study. *BMJ*. 2017;357:j1456. doi:10.1136/BMJ.J1456
2. Cradock A, Barret J, Hull T, Fields B. Evidence to Inform a Cycling and Walking Investment Strategy.; 2019. <https://cdn1.sph.harvard.edu/wp-content/uploads/sites/84/2019/05/Evidence-to-Inform-a-Cycling-and->

Walking-Investment-Strategy_2019_04_30.pdf.

3. Buehler R. Determinants of bicycle commuting in the Washington, DC region: The role of bicycle parking, cyclist showers, and free car parking at work. *Transp Res Part D Transp Environ.* 2012;17(7):525-531. doi:10.1016/j.trd.2012.06.003
4. Wahlgren L, Schantz P. Exploring bikeability in a suburban metropolitan area using the Active Commuting Route Environment Scale (ACRES). *Int J Env Res Public Heal.* 2014;11(8):8276-8300. doi:10.3390/ijerph110808276
5. Transportation Alternatives. Indoor Bicycle Parking. <https://www.transalt.org/issues/bike/network/parking/indoor>.
6. Transportation Alternatives. Bicycle Parking Solutions: A Resource Guide for Improving Secure Bicycle Parking in New York City. <https://www.transalt.org/sites/default/files/issues/bike/bikeparking.pdf>.
7. Chapman R, Howden-Chapman P, Keall M, et al. Increasing active travel: Aims, methods and baseline measures of a quasi-experimental study. *BMC Public Health.* 2014;14(1). doi:10.1186/1471-2458-14-935
8. King AC, Powell KE, Physical Activity Guidelines Advisory Committee, Committee PAGA. 2018 Physical Activity Guidelines Advisory Committee Scientific Report. 2018. <https://health.gov/paguidelines/second-edition/report.aspx>.
9. Pucher J, Dill J, Handy S. Infrastructure, programs, and policies to increase bicycling: an international review. *Prev Med.* 2010;50 (suppl:S106-25. doi:10.1016/j.ypmed.2009.07.028
10. Wardman M, Tight M, Page M. Factors influencing the propensity to cycle to work. *Transp Res Part A Policy Pract.* 2007;41(4):339-350. doi:https://doi.org/10.1016/j.tra.2006.09.011
11. National Association of City Transportation Officials. Urban Bikeway Design Guide, 2nd ed. 2014:260. <http://nacto.org/publication/urban-bikeway-design-guide/>.
12. Reynolds CC, Harris MA, Teschke K, Cripton PA, Winters M. The impact of transportation infrastructure on bicycling injuries and crashes: a review of the literature. *Environ Heal.* 2009;8(1):47. doi:10.1186/1476-069X-8-47
13. Pucher J, Buehler R. Safer cycling through improved infrastructure. *Am J Public Health.* 2016;106(12):2089-2091. doi:10.2105/AJPH.2016.303507
14. Walk Score. Bike Score. <https://www.walkscore.com/bike-score-methodology.shtml>. Accessed January 1, 2018.
15. U.S. Green Building Council. LEED BD+C: New Construction - Bicycle Facilities. <https://www.usgbc.org/credits/new-construction-data-centers-new-construction-hospitality-new-construction-warehouse-and--0?return=/credits/New%20Construction/v4.1https://www.usgbc.org/credits/new-construction-data-centers-new-construction-hospitality-new-construction-warehouse-and--0?return=/credits/New%20Construction/v4.1>. Accessed January 1, 2020.
16. U.S. Green Building Council. LEED BD+C: Core and Shell - Surrounding density and diverse uses. <https://www.usgbc.org/credits/core-shell/v2012/ltc4>. Accessed January 1, 2018.
17. U.S. Green Building Council. LEED BD+C: Retail - Bicycle Facilities. <https://www.usgbc.org/credits/retail-new-construction/v4-draft/ltc6>. Accessed January 1, 2018.